

UX Requirements — Quick Measure Mapping & Visibility Controls

Author: UX Product Designer | **Last updated:** April 2026

Scope: Two workflows within the 2D Takeoff app: (1) Quick Measure → Key Measure mapping, (2) visibility controls.

1. Quick Measure → Key Measure Mapping

1.1 User Goal

Organize rough on-canvas measurements into the project's formal cost-code structure, without leaving the measurement list context.

1.2 Entry Points

#	Trigger	Where	When visible	Use case	Notes
E1	"Assign" button	Right panel, inline on measurement row	On row hover (replaces dimensions)	Assign a single item during line-by-line review	
E2	"Assign Selected" button	Right panel, bulk action bar	When ≥ 1 checkbox is checked	Assign a batch to the same Key Measure + Section	

1.3 Selection & Bulk Actions

#	Element	Behavior	Notes
S1	Row checkbox	Toggles individual measurement into/out of selection	
S2	"Select All" toggle	Selects or deselects all visible measurements	Label switches between "Select All" / "Deselect All"
S3	Bulk action bar	Slides in above list when ≥ 1 item selected	Shows: "N selected", clear (x), "Assign Selected", delete icon
S4	Post-assignment clear	All checkboxes deselected after successful apply	

1.4 Assignment Popover

#	Aspect	Requirement	Notes
A1	Format	Popover (not modal) — measurement list stays visible behind it	Keeps spatial context
A2	Positioning	Anchored near trigger button; prefers left side, flips if no room; clamped to viewport	
A3	Dismiss	Click outside to close; Cancel button; × in header	No confirmation needed — action is non-destructive
A4	Context summary	Shows count + names of measurements being assigned	Truncated if long
A5	Key Measure field	Required. Searchable dropdown. Each option shows name + category. Selected state shows name + clear (×)	
A6	Section field	Optional. Same searchable dropdown pattern	Can be left empty
A7	Apply button	Disabled until Key Measure is selected. No “are you sure” step	
A8	Success feedback	Toast notification; popover closes; selection cleared; left panel refreshes	

1.5 Re-assignment Rules

#	Scenario	Behavior	Notes
R1	Assign to a new Key Measure	Creates additional mapping (many-to-many)	A measurement can belong to multiple Key Measures
R2	Assign to same Key Measure again	Upserts — updates Section if changed, no duplicate created	

1.6 Post-Assignment Feedback

#	Location	Indicator	Interaction	Notes
F1	Right panel row	Green link icon (↔)	Single KM: hover highlights, click pins/unpins. Multiple KMs: click opens popover (see F4)	Badge shows count when > 1
F2	Left panel row	Green link icon (↔)	Hover shows tooltip with measurement name	Only for assignments originating from Quick Measures panel
F3	Cross-panel	Highlighting targets a specific assignment row in the left panel, not all rows for that measurement	Emerald tint + ring on the single targeted row	

1.7 Multi-Assignment Link Popover

When a Quick Measure is assigned to **multiple** Key Measures, the link icon behavior changes from a simple hover to an interactive popover.

#	Aspect	Requirement	Notes
F4	Trigger	Click the link icon (which shows a count badge)	Hover alone does not open the popover for multi-assignment items
F5	Popover content	“Assigned to” header + a row for each Key Measure	Each row has a Layers icon + KM name
F6	Hover within popover	Hovering a KM row highlights that specific assignment row in the left panel	Only one row highlighted at a time; user can scan through them
F7	Click within popover	Clicking a KM row pins the highlight and closes the popover	Highlight persists while user moves their attention to the left panel
F8	Dismiss	Click outside the popover clears the highlight and closes it	
F9	Problem solved	User no longer loses the highlight when moving from the link icon to the left panel	Pinning is the key mechanism — highlight survives the mouse move

2. Visibility Controls

2.1 User Goal

Manage visual clutter on a dense takeoff sheet — hide individual items for focused work, or sweep away entire categories temporarily — without losing track of what exists.

2.2 Two-Layer Model

Layer	Scope	Persists?	Control surface	Notes
Per-item toggle	One measurement	Yes (saved to DB)	Eye icon in side panels	User's deliberate intent
Canvas filters	Groups by source / type / section	No (session only, resets on reload)	Legend panel on canvas	Temporary viewing aid

A measurement is drawn on canvas only when it passes **both** layers.

2.3 Per-Item Visibility Toggle

#	Aspect	Requirement	Notes
V1	Locations	Eye icon on every row in left panel + right panel; also in right panel's ellipsis menu ("Hide"/"Show")	
V2	Feedback	Optimistic — icon changes immediately, DB update in background	
V3	Works during filtering	Toggling works even on items hidden by canvas filters	Lets user pre-set visibility for when filter is cleared

Visual States

#	Condition	Eye icon	Row treatment	Tooltip	Notes
V4	Visible	Eye, colored to match measurement	Normal	“Hide markup”	
V5	Hidden by user	EyeOff, gray	Normal	“Show markup”	
V6	Hidden by canvas filter	EyeOff, gray	Entire row at 45% opacity	“Hidden by canvas filter”	Regardless of user toggle state

2.4 Canvas Filter Legend

#	Aspect	Requirement	Notes
L1	Location	Bottom-right corner of canvas	Out of the way of blueprint content
L2	Collapsed state	Pin icon + “Legend” + chevron; shows N/M count if filters active	Persistent reminder that filtering is on
L3	Sections	Divided into: Source, Type, Section, Footer	Separated by dividers

Source Filters (show/hide toggles)

All default to **ON**. Independent — any combination valid. AND logic (must pass all to appear on canvas).

#	Filter	What it controls	Default	Icon	Notes
L4	Key Measures	Measurements with at least one assignment	ON	Layers	
L5	Quick Measures	Measurements with no assignments	ON	Eye	
L6	Auto-Takeoff	Measurements where isAI = true	ON	Sparkles	

Visual treatment:

State	Icon	Label	Row	Design rationale	Notes
ON (default)	Eye, subtle gray	Normal	Quiet, neutral	Default state shouldn't demand attention	
OFF	EyeOff, faded	Strikethrough, dimmed	Draws attention	User's eye goes to what's different from normal	

Type Filters (inclusive select)

Default: **nothing selected = all shown**. Selecting types restricts canvas to those types only.

#	Type	Color dot	Icon	Notes
L7	Area	Green (#10B981)	Square	
L8	Linear	Orange (#F97316)	Dash	
L9	Count	Amber (#F59E0B)	Circle	

Visual treatment: Ring/dot indicator (not eye icon) — different affordance signals different semantics (“filter to” vs. “show/hide”).

State	Indicator	Label	Notes
Not selected (default)	Empty ring	Normal	
Selected	Filled gray dot	Bold	

Section Filter

#	Aspect	Requirement	Notes
L10	Format	Dropdown multi-select with checkboxes	
L11	Default	No selection = all sections shown	Same inclusive logic as Type
L12	Quick actions	“Select All” / “Clear” at bottom of drop-down	
L13	Count	Each section shows its measurement count	

Footer

#	Aspect	Requirement	Notes
L14	Visibility	Only shown when filters are active	
L15	Content	“N of M visible” + Reset button	Reset returns all filters to defaults

2.5 Side Panels During Filtering

#	Requirement	Rationale	Notes
P1	Panels always show the full measurement list	Removing items would lose context; user might think items were deleted	
P2	Filtered-out items are dimmed (45% opacity)	Visual cue that items exist but aren't on canvas right now	
P3	Toggles still work on dimmed items	User can pre-set visibility for when filter is removed	

3. Hover Highlighting

#	Trigger	Canvas behavior	Panel behavior	Notes
H1	Hover measurement row in either panel	Spotlights that markup — full color, thick stroke, floating label (name + value + unit)	Hovered row gets blue tint	
H2	(same)	All other markups fade to uniform low-opacity gray		
H3	Hover hidden measurement	Canvas temporarily reveals it		Lets user identify which area a hidden measurement covers
H4	Hover ends	Canvas returns to normal state		

4. State Matrix (QA Reference)

User toggle	Canvas filter	Canvas	Panel row	Eye icon	Notes
Shown	Passes	✓ Drawn	Full opacity	Colored eye	
Shown	Filtered out	✗ Not drawn	45% opacity	Gray eye-off	Tooltip: "Hidden by canvas filter"
Hidden	Passes	✗ Not drawn	Full opacity	Gray eye-off	Tooltip: "Show markup"
Hidden	Filtered out	✗ Not drawn	45% opacity	Gray eye-off	Tooltip: "Hidden by canvas filter"
Any	Any + hovered	✓ Temp drawn	Highlighted	Unchanged	

5. Design Decisions Log

#	Decision	Rationale	Notes
D1	Popover, not modal, for assignment	Keeps measurement list visible; lighter interaction	
D2	Anchor popover near trigger	Reduces eye travel; spatial context	
D3	Inline “Assign” on hover, not always visible	Keeps rows clean when not assigning	
D4	Two independent visibility layers	Per-item = persistent intent; filters = temporary aid; conflating creates confusion	
D5	Filtered items dimmed, not removed	Prevents “where did my items go”; allows pre-toggling	
D6	Quiet default state in legend	All filters ON by default — highlighting the default wastes attention	
D7	Eye icon for source toggles, dot for type	Different semantics deserve different affordances	
D8	Hover reveals hidden items on canvas	Lets user identify spatial location without toggling	
D9	Filters don’t persist across sessions	Filters are a temporary lens, not saved configuration	
D10	No confirmation on assign	Non-destructive action; can be re-done or overwritten	